Basics of Autovacuum

Automatic Vacuum, or *Autovacuum*, is a background process that runs by default when PostgreSQL starts. Autovacuum can be tuned using a variety of parameters. On modern production servers the default Autovacuum parameter values are considered to be very conservative. For this reason, Autovacuum tuning guides often recommend changing the values to make Autovacuum run more frequently.

If you're completely new to Autovacuum, it can be thought of as the "garbage collection process"[2] for PostgreSQL. In a roughly similar way to how garbage collection runs periodically in garbage-collected programming languages when objects are deallocated, Autovacuum runs periodically for dead row versions in a database that are no longer needed for any transactions.

Visible Rows are made up of row versions behind the scenes. Row versions can be liv,e reflecting what's currently visible to all live and running transactions, or dead. When a row is updated or deleted in PostgreSQL it's not actually updated or removed physically, but instead a new row version called a tuple reflects the changes from the update or delete.

Run the following query from psql. This query shows the `ctid` column, which is a hidden column. The two values displayed are the page number and tuple number.[3]

```
SELECT ctid,id FROM users WHERE id = 1;
```

Update the same User by running `UPDATE users SET first_name = 'Jane' WHERE id = 1;`.

Now run the same query showing the `ctid` from above. The tuple version will change. The page number may be the same unless a lot of updates have happened, and in that case PostgreSQL may place the row version in a new page.

When a former row version is no longer referenced by any live transaction, it becomes dead. At that point the dead row version may be removed. Until it is removed, dead row versions are commonly referred to as *bloat*.

The way Autovacuum works in depth and the dozens of tunable parameters are beyond our scope. The basic way it works and a couple of tuning parameters will help you get started.

2. https://www.postgresql.org/docs/current/routine-vacuuming.html
3. https://nidhig631.medium.com/ctid-field-in-postgresql-d26977de7b58